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Top Skills

AI System Architecture
Solution Architecture
Generative AI Systems

Languages

English
Hindi

Certifications

Microsoft Certified: Azure
Fundamentals
AHIP Healthcare 101
Microsoft Certified: Azure AI
Fundamentals

Mohammad Talah Sheikh

Solution Designer | Senior AI & Full-Stack Engineer | Lead Engineer
| Enterprise-Scale GenAI & Distributed Systems | Java, Node.js,
React

Greater Sydney Area

Summary

I am a Senior Technical Lead, Solution Designer, and AI-First Full-Stack Engineer with 15+ years of experience building enterprise-grade systems at the intersection of architecture, scalability, and Generative AI.

I bring the engineering maturity of a decade in Java and microservices into modern AI systems—ensuring that GenAI solutions are not just innovative, but secure, observable, and production-ready. For the past two years, I have intentionally pivoted into an AI-first engineering mindset, focusing on enterprise GenAI platforms within the Azure AI ecosystem.

From Java Lead to AI Solutions Engineer

Over 12 years, I built mission-critical platforms for global financial organisations including Westpac and Fiserv, where reliability, performance, and regulatory compliance were non-negotiable. This foundation allows me to approach AI differently: not as experimentation, but as enterprise software engineering.

My recent work centres on the shift from imperative services to agentic AI systems—goal-driven architectures capable of reasoning, tool usage, and autonomous task execution within controlled enterprise boundaries.

AI Specialisation (2023–Present)

Agentic Orchestration — Designing multi-agent systems using LangChain and LangGraph, incorporating stateful workflows, reflection loops, task decomposition, and long-term memory.
Production-First RAG — Architecting Retrieval-Augmented Generation pipelines using Azure AI Search and vector databases,

with strong focus on hybrid retrieval, chunking strategies, and hallucination control in financial contexts.

Asynchronous AI Systems — Building resilient AI backends using Kafka, Redis, and queue-based orchestration, enabling safe handling of high-latency LLM workloads and failure-tolerant execution.

The Foundation

My background in Spring Boot, microservices, cloud-native architecture, and distributed systems ensures that the “AI brain” is always connected to a resilient and compliant “body.” I regularly bridge legacy enterprise systems and modern LLM-based platforms, including recognition for modernising core payment systems with AI-ready API layers.

I am passionate about moving GenAI beyond chat interfaces and embedding it directly into business workflows—building systems that can reason, invoke tools, and deliver measurable ROI at enterprise scale.

Core Stack: Azure OpenAI • LangGraph • Python • Node.js • Spring Boot • React • Kafka • Vector Databases

Experience

Cognizant

Senior Consulting Manager

May 2026 - Present (1 month)

Australia

Data & AI Architect

Fiserv

6 years 8 months

Lead Engineer

March 2024 - April 2026 (2 years 2 months)

Client Engagement: St.George Bank (Westpac Group)

Senior Software Engineer

Mar 2024 – Present

Program: Compass Digital Banking Platform

- Currently working as Senior Engineer on the Compass platform, supporting St.George's digital banking capabilities.
- Designing and delivering secure, high-performance APIs integrated with Westpac Group platforms.
- Contributing to ongoing platform modernisation and stability initiatives, with focus on performance, observability, and reliability.
- Collaborating with architects and senior stakeholders on design decisions and technical improvements.

AI & Generative AI Engineering (Self-Driven / Ongoing)

January 2023 - April 2026 (3 years 4 months)

Sydney, NSW

AI & Generative AI Engineering (Self-Driven / Ongoing)

2023 – Present

- Actively building expertise in Generative AI and AI-first system design through structured self-learning and hands-on projects.
- Designed and implemented small AI-driven applications to explore:
 - Retrieval-Augmented Generation (RAG)
 - Agent-based orchestration and workflows
 - LLM integration within backend systems
- Currently developing an end-to-end AI-driven platform to demonstrate:
 - AI orchestration
 - Backend integration
 - Scalable system design
 - Production-ready architecture

Lead Engineer

August 2023 - March 2024 (8 months)

Sydney, New South Wales, Australia

Client Engagement: Westpac Banking Corporation

Senior Engineer → Solution Designer

Aug 2023 – Mar 2024

Program: Open Banking / API Modernisation

- Transitioned into a solution-oriented role, acting as Senior Engineer with Solution Designer responsibilities.

- Collaborated with enterprise architects and business stakeholders to analyse production issues and design high-level solutions for Open Banking APIs.
- Owned low-level solution design, working across teams to align API contracts, integration flows, and test strategies.
- Led SOAP-to-REST migrations, improving interoperability, maintainability, and readiness for modern API consumers.
- Ensured all solutions met strict banking NFRs through performance testing, profiling, and design validation.

Lead Engineer

August 2023 - September 2023 (2 months)

Sydney, New South Wales, Australia

Solution Designer & Lead Engineer (POC)

Aug 2023

Project: Galaxy Payment Switch – API Enablement (HP Tandem Mainframe)

- Solutioned and delivered an end-to-end proof-of-concept to expose a real-time REST service layer over the Galaxy payment switch (HP Tandem mainframe).
- Designed a secure API abstraction enabling customers to perform actions without reliance on contact centres or manual mainframe job execution.
- Implemented direct integration with underlying mainframe systems to retrieve and process card-related data.
- Demonstrated how legacy payment infrastructure could be modernised using API-first, event-ready architecture.
- POC was successfully approved for productisation as a new Galaxy platform capability.
- Received awards and formal recognition for innovation, ownership, and technical excellence.

Lead Engineer

September 2019 - July 2023 (3 years 11 months)

Sydney, New South Wales, Australia

Client Engagement: Westpac Banking Corporation

Senior Software Engineer / Lead Engineer

Sep 2019 – Jul 2023

Program: Westpac Digital Platforms (WDP)

- Joined as Senior Software Engineer and progressed into a Lead Engineer role, contributing to and leading delivery of enterprise-scale digital banking services.
- Designed, developed, and maintained high-volume REST APIs and integration services supporting customer-facing banking journeys.
- Led technical design and implementation across multiple components within complex integration flows on Westpac's WDP platform.
- Modernised legacy Java services by migrating servlet-based and SOAP services to Spring Boot microservices, upgrading JVM versions (Java 7 → 8 → 11).
- Worked closely with architects, business analysts, QA, and platform teams to deliver secure, performant, and compliant solutions.
- Performed production troubleshooting, performance profiling, and optimisation using Splunk and custom diagnostics.
- Actively mentored engineers and supported QA teams in adopting automation-first engineering practices.

Infosys

8 years 10 months

Technical Lead

February 2017 - August 2019 (2 years 7 months)

Sydney, New South Wales, Australia

Client Engagement: Westpac Banking Corporation

Program: Customer Service Hub – Mortgage Origination & Servicing Platform

- Led engineering design and delivery for functional and automation platforms supporting a large-scale mortgage origination and servicing system.
- Owned architecture and development of Java-based automation frameworks, treating automation as a first-class engineering system rather than a testing utility.
- Worked closely with solution architects, backend developers, and business stakeholders to ensure automation aligned with core system design, integration flows, and business rules.
- Acted as technical authority for complex end-to-end workflows, including loan origination, settlement, and downstream integrations.
- Led and mentored a 15-member offshore engineering team, driving coding standards, design practices, and delivery quality.
- Enabled functional engineers to transition into automation engineers through structured training in Java, design principles, and framework usage.
- Contributed to integration analysis, dependency mapping, and defect root-cause analysis, often identifying design gaps in upstream services.

Technical Analyst

November 2010 - January 2017 (6 years 3 months)

Payment Master – Blue Cross Blue Shield of Minnesota

- Designed and maintained payment processing services, supporting both legacy SOAP and modern REST interfaces.
- Developed backend Java services and UI components (JSP) for customer payment workflows.
- Built comparison engines and reporting modules supporting financial reconciliation.
- Conducted code reviews and mentored junior engineers and automation testers.
- Progressed from Junior Developer to Technical Lead, demonstrating consistent technical and leadership growth.

Education

Technocrats institute of technology, bhopal

Bachelor of Engineering (BE), Computer Science · (2006 - 2010)

Vatsalya sn sec school, sagar